

DeSurv Factor Diagnostics: Convergence, Survival Associations, and Gene Programs

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[v2] Raw gene intersection: 17900 genes across TCGA_PAAD + CPTAC
[v2] Per-platform z-standardization (colMean=0, colSD=1 per cohort) ...
[v2] Quantile normalising merged matrix (273 x 17900) ...
[v2] Genes retained after top-2000 variance filter: 2000
[v2] Skipping rank transform (rank_transform = FALSE).

[v2] Raw gene intersection: 17900 genes across TCGA_PAAD + CPTAC
[v2] Per-cohort gene selection (top-3000, method=combined_rank) before normalization ...
[v2] Genes after per-cohort selection and intersection: 2064
[v2] Per-platform z-standardization (colMean=0, colSD=1 per cohort) ...
[v2] Skipping normalization (log-transform only; 273 x 2064) ...
[v2] Skipping rank transform (rank_transform = FALSE).
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1. Configuration Overview

Table 1: Summary of all five configurations. K_{eff} = factors with $|| > 0.001$. Betas reported after sign correction (so positive = adverse prognosis). RMSE = final training reconstruction error. Mean ext. C = mean concordance across 5 held-out PDAC cohorts.

ID	Label	Model	K	K_{eff}	Active $\hat{\beta}$ (by factor)	Iters	RMSE	Mean ext. C
D1	LB orig (M4)	LB	3	1	k3: -0.482	16	1.0157	0.616
D2	YFB orig (M5)	YFB	3	2	k1: -0.053; k2: +0.026	19	1.0252	0.624
D3	LB DeSurv-aligned	LB	7	2	k2: -0.439; k3: +0.205	105	0.8671	0.622
D4	YFB DeSurv-aligned	YFB	7	2	k3: +0.011; k7: -0.041	13	0.9330	0.636
D5	YFB DeSurv + cohort	YFB	8	2	k7: -0.028; k8: +0.010	52	0.8669	0.614

2. Convergence Traces

3. Survival Coefficients

4. Per-Configuration Factor Analysis

D3 — LB DeSurv-aligned

K=7, K_{eff} =2 | Converged in 105 iterations | Final RMSE = 0.8671 | Mean ext. C = 0.622

Factor summary:

Factor	$\hat{\beta}$	Status	Log-rank p	PVE (%)
1	0.0000	—	0.5694	16.12
2	-0.4394	active	0.0095	7.64
3	0.2051	active	0.1201	17.68
4	0.0000	—	NA	0.00
5	0.0000	—	0.5733	6.33
6	0.0000	—	0.9517	23.44

Factor	$\hat{}$	Status	Log-rank p	PVE (%)
7	0.0000	—	NA	0.00

Kaplan-Meier stratification (training cohort, n=273) — 2 active factor(s):

Top 15 genes per active factor (by |posterior weight|):

Factor 2 — $\hat{} = -0.4394$ (protective), PVE = 7.64%, log-rank $p = 0.0095$

Gene	Posterior weight
MET	1.04296
ITGA2	0.94124
ITGA3	0.93744
CDCP1	0.90915
ANLN	0.90835
EGFR	0.90044
PGK1	0.89091
ASPH	0.88957
AP1S3	0.88534
FAM3C	0.88468
RASA1	0.87624
ASAP2	0.87481
TMOD3	0.86808
SLC2A1	0.86644
BCAR3	0.86397

Factor 3 — $\hat{} = +0.2051$ (adverse), PVE = 17.68%, log-rank $p = 0.1201$

Gene	Posterior weight
CAPN5	1.14815
PLEKHA6	1.05733
FMO5	1.05348
TJP3	1.04867
C9orf152	1.04318
MLPH	1.04290
KCNE3	1.03135
PIP5K1B	1.02971
CAPN8	1.01895
PLA2G10	1.01821
TSPAN3	1.01788
FUT2	1.00512
LPCAT4	1.00287
RBM47	0.99375
BCAS1	0.98665

D4 — YFB DeSurv-aligned

K=7, K_eff=2 | Converged in 13 iterations | Final RMSE = 0.9330 | Mean ext. C = 0.636

Factor summary:

Factor	$\hat{}$	Status	Log-rank p	PVE (%)
1	0.0000	—	NA	0.00
2	0.0000	—	NA	0.00
3	0.0113	active	0.0123	2.92
4	0.0000	—	NA	0.00
5	0.0000	—	0.3850	2.95
6	0.0000	—	0.3645	2.88
7	-0.0408	active	0.0003	1.90

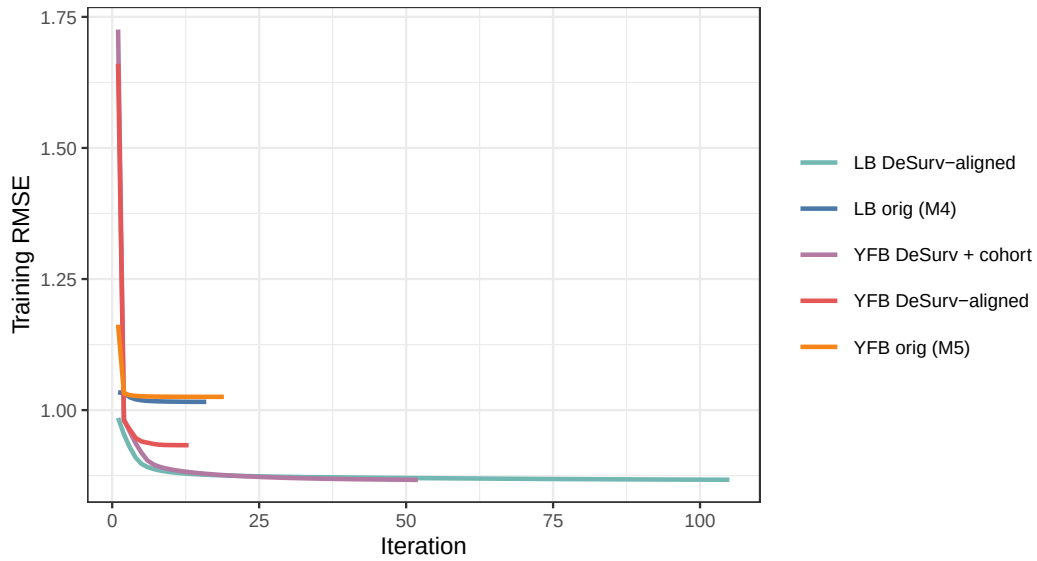


Figure 1: Training RMSE by iteration for all five configurations. Each line represents one configuration; convergence is declared when the relative ELBO change falls below $1e-5$. DeSurv-aligned LB (D3) takes the most iterations to converge; YFB models with DeSurv preprocessing converge rapidly.

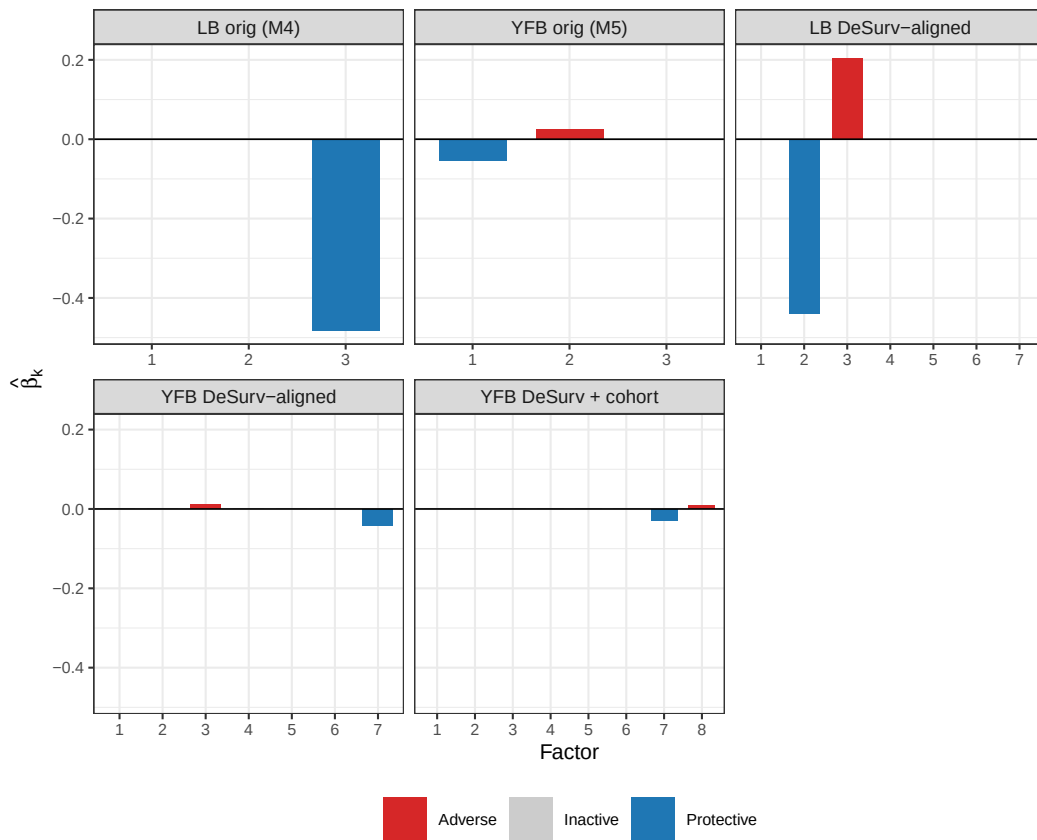
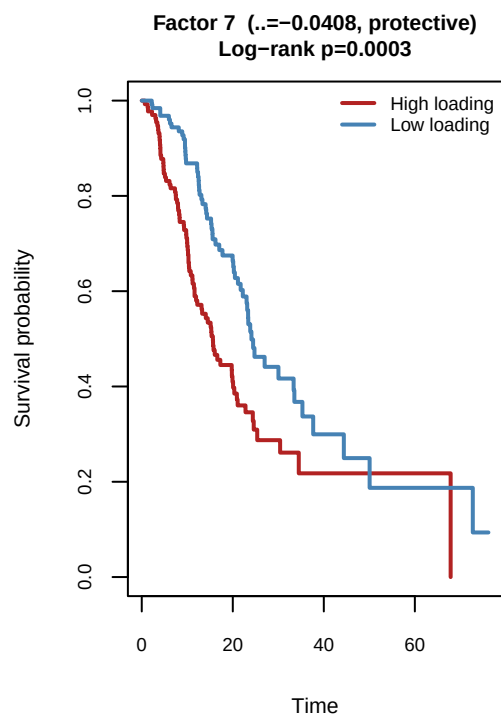
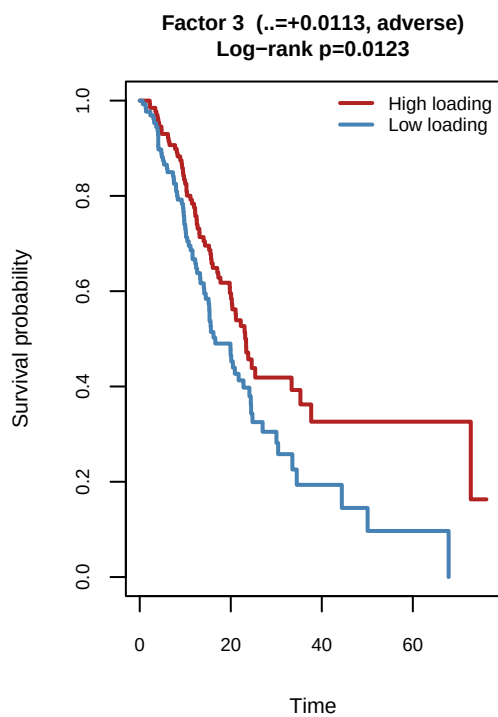
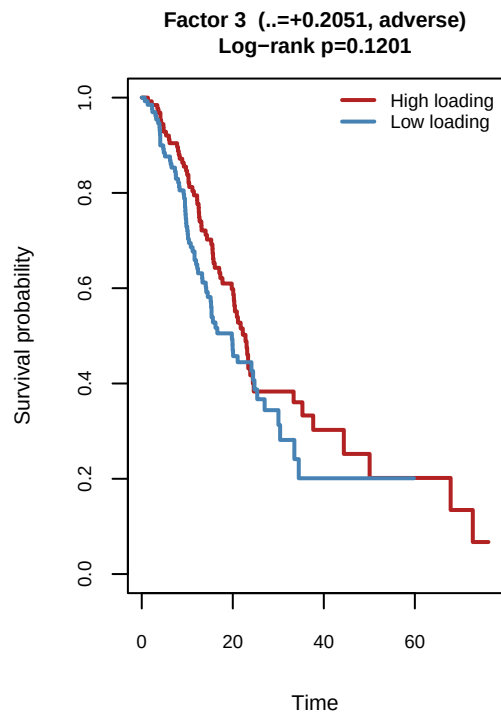
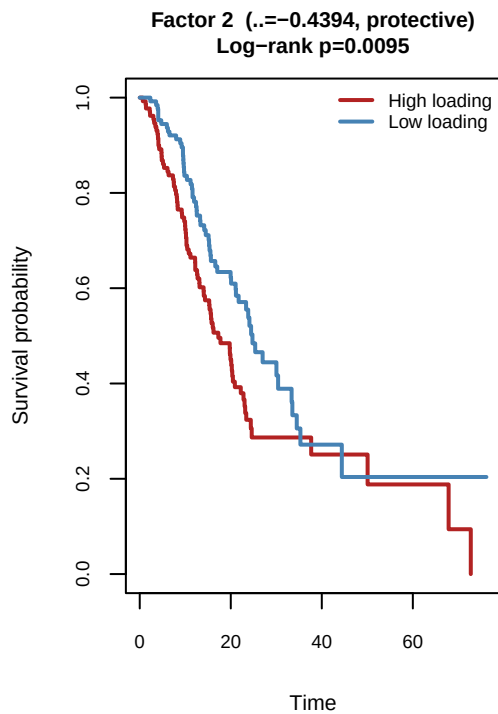


Figure 2: Posterior mean survival coefficients $\hat{\beta}_k$ for each factor within each configuration. Active factors ($|\hat{\beta}_k| > 0.001$) are coloured; inactive are grey. Positive = adverse prognosis (high loading $\rightarrow$ worse survival); negative = protective. YFB values are on a smaller scale (~ 0.01 – 0.05) than LB (~ 0.2 – 0.5) because YFB normalises factor columns, compressing the coefficient scale.



Kaplan-Meier stratification (training cohort, n=273) — 2 active factor(s):

Top 15 genes per active factor (by |posterior weight|):

Factor 3 — $\hat{\beta} = +0.0113$ (adverse), PVE = 2.92%, log-rank $p = 0.0123$

Gene	Posterior weight
CAPN5	2.63664
LPCAT4	2.62750
MLPH	2.54719
SLC45A3	2.46790
TJP3	2.44992
PLEKHA6	2.39411
MTMR11	2.38587
GALNT6	2.38010
CAPN8	2.32556
TSPAN3	2.32296
CYP2S1	2.32198
PAK1	2.31960
FAM83E	2.30680
KCNK1	2.30463
KCNE3	2.30441

Factor 7 — $\hat{\beta} = -0.0408$ (protective), PVE = 1.90%, log-rank $p = 0.0003$

Gene	Posterior weight
ITGA3	2.58918
MET	2.55010
BCAR3	2.48200
FAM3C	2.44707
GAPDH	2.40522
ENO1	2.40133
TPI1	2.32039
PGK1	2.30881
CDCP1	2.29102
LDHA	2.28484
SLC16A3	2.26966
ITGB4	2.24910
SLC2A1	2.24869
TGM2	2.21551
S100A2	2.21543

D5 — YFB DeSurv + cohort

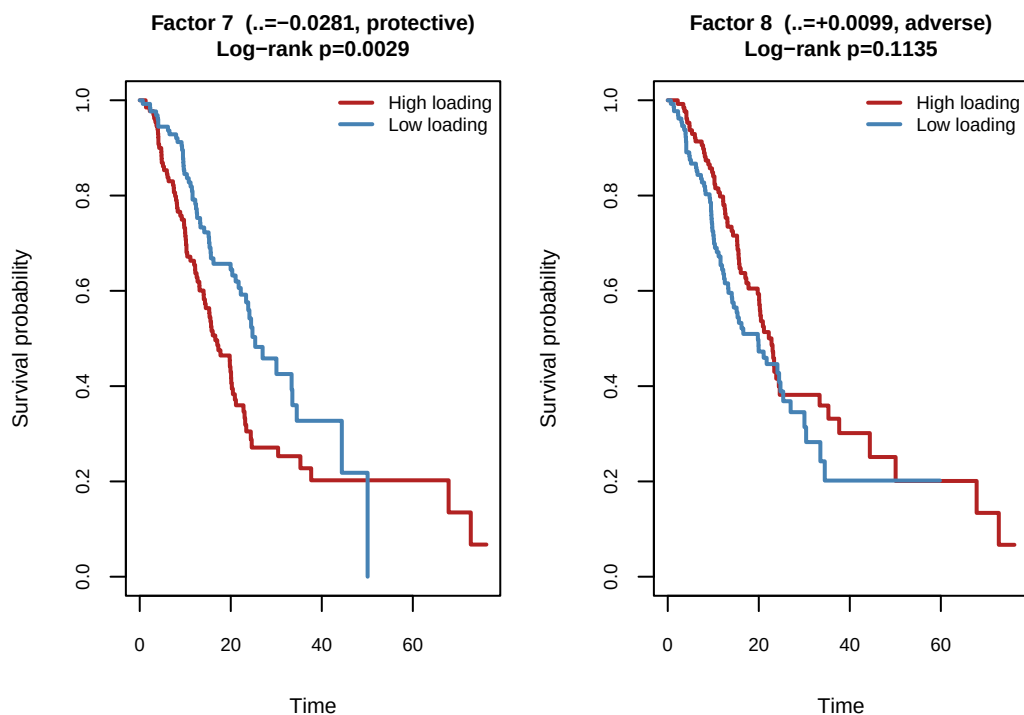
K=8, K_eff=2 | Converged in 52 iterations | Final RMSE = 0.8669 | Mean ext. C = 0.614

Factor summary:

Factor	$\hat{\beta}$	Status	Log-rank p	PVE (%)
1	0.0000	—	NA	0.00
2	0.0000	—	NA	0.00
3	0.0000	—	NA	0.00
4	0.0000	—	NA	0.00
5	0.0000	—	0.9492	36.08
6	0.0000	—	0.2541	13.43
7	-0.0281	active	0.0029	12.65
8	0.0099	active	0.1135	17.14

Kaplan-Meier stratification (training cohort, n=273) — 2 active factor(s):

Top 15 genes per active factor (by |posterior weight|):



Factor 7 — $\hat{\beta} = -0.0281$ (protective), PVE = 12.65%, log-rank $p = 0.0029$

Gene	Posterior weight
ITGA3	2.54520
SLC16A3	2.48983
FSCN1	2.38915
SLC2A1	2.38543
PLAU	2.38268
MMP14	2.35920
ANXA1	2.31564
LAMB3	2.31383
P4HA1	2.28331
ALDOA	2.28205
KRT19	2.28175
CDCP1	2.27363
LDHA	2.25637
ITGB4	2.24328
MET	2.21003

Factor 8 — $\hat{\beta} = +0.0099$ (adverse), PVE = 17.14%, log-rank $p = 0.1135$

Gene	Posterior weight
CAPN5	1.55063
MLPH	1.42236
TJP3	1.41188
SYTL2	1.40946
KIAA1211	1.39554
LPCAT4	1.38417
PLEKHA6	1.37616
PLA2G10	1.37444
C9orf152	1.36992
CAPN8	1.36946
TSPAN3	1.36449
FUT2	1.34713
KALRN	1.34462
BCAS1	1.34323
KCNE3	1.34278

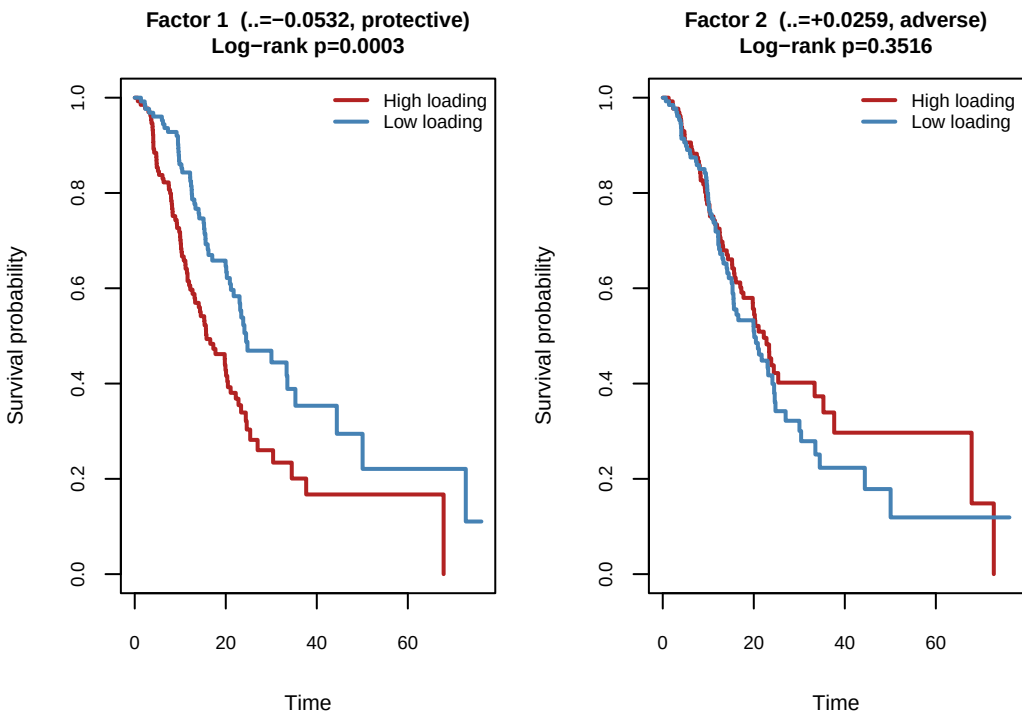
D2 — YFB orig (M5)

K=3, K_eff=2 | Converged in 19 iterations | Final RMSE = 1.0252 | Mean ext. C = 0.624

Factor summary:

Factor	$\hat{}$	Status	Log-rank p	PVE (%)
1	-0.0532	active	0.0003	0.93
2	0.0259	active	0.3516	1.05
3	0.0000	—	0.3132	0.37

Kaplan-Meier stratification (training cohort, n=273) — 2 active factor(s):



Top 15 genes per active factor (by |posterior weight|):

Factor 1 — $\hat{} = -0.0532$ (protective), PVE = 0.93%, log-rank $p = 0.0003$

Gene	Posterior weight
LDHA	2.58444
SLC2A1	2.55696
ADM	2.45985
ALDOA	2.29922
NDRG1	2.16013
ALDOC	2.15481
PTTG1IP	2.11143
EGLN3	2.06470
APLN	2.06385
PTGES	2.05615
FGF23	2.04261
HK2	2.00464
PDK1	1.98567
APOL1	1.97216
TNNT1	1.96028

Factor 2 — $\hat{} = +0.0259$ (adverse), PVE = 1.05%, log-rank $p = 0.3516$

Gene	Posterior weight
ASPHD2	2.86254
LPCAT4	2.81252
FHDC1	2.68463
OSBPL2	2.63090
KCNK1	2.62386
RETSAT	2.57970
PAK1	2.57536
CYP2S1	2.48637
SH2D4A	2.44926
NQO1	2.36951
LDLR	2.35274
KPNA7	2.28373
BFAR	2.27430
AKAP1	2.24522
ADAP1	2.24416

5. GEP Loading Heatmaps

For each DeSurv-aligned configuration, the heatmap shows posterior mean gene weights $\hat{F}_{jk}$ for the top 40 genes (by maximum $|\hat{F}_{jk}|$ across all factors). Rows are genes; columns are factors. Colour encodes the weight (blue = negative / low expression associated with high loading, red = positive).

Asterisked factor labels (*) mark active factors ($|r| > 0.001$). The column width and visual complexity increase with K ; for D3/D4 ($K=7$) and D5 ($K=8$), ARD prunes most factors to near-zero, leaving a sparse set of biologically relevant programs.

6. D4 vs D5 Active Gene Overlap

Both D4 and D5 use the same 2,064-gene input set. Here we compare the top-50 genes in each active factor between configurations to assess whether the cohort indicator changes what biological programs are identified.

Table 14: Gene overlap between D4 and D5 active factors (top-50 genes by $|EF|$ weight). Jaccard = shared / union. High overlap indicates the cohort indicator changed the survival weighting but not the gene program structure.

D4 factor	D5 factor	Shared top-50 genes	Jaccard
D4 F3 (=+0.0113)	D5 F7 (=-0.0281)	0	0.000
D4 F3 (=+0.0113)	D5 F8 (=+0.0099)	39	0.639
D4 F7 (=-0.0408)	D5 F7 (=-0.0281)	28	0.389
D4 F7 (=-0.0408)	D5 F8 (=+0.0099)	0	0.000

7. Proportional Hazards Check (Active Factors)

D3 — LB DeSurv-aligned:

Term	χ^2	df	p
F2	1.5489	1	0.2133
F3	1.1199	1	0.2900
GLOBAL	3.9940	2	0.1357

Global PH $p = 0.1357$ — no evidence of a PH violation.

D4 — YFB DeSurv-aligned:

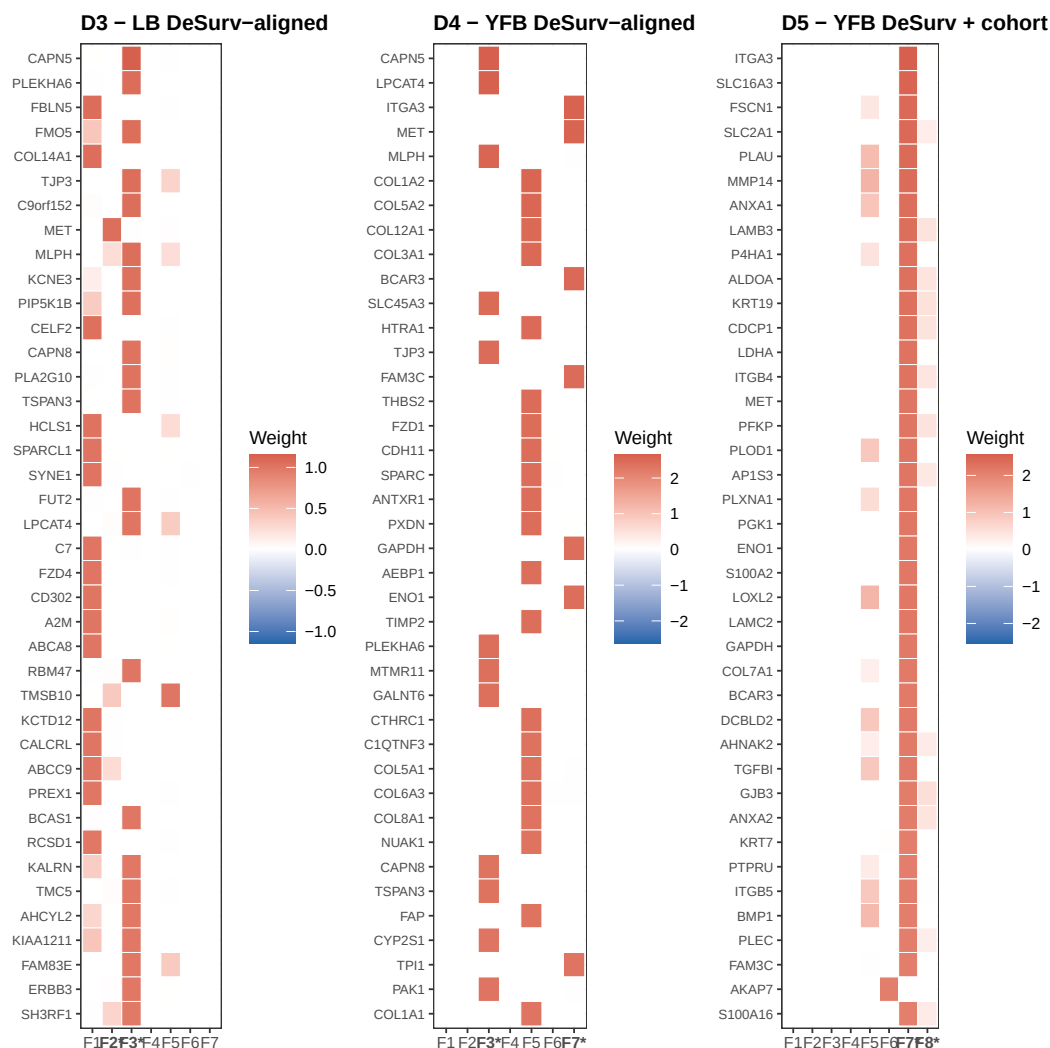


Figure 3: GEP loading heatmaps for D3, D4, and D5 (top 40 genes by max absolute weight across all factors). Columns = factors; rows = genes. Active factors ($| | > 0.001$) are underlined in the column labels. Each column is the molecular signature of one gene expression program.

Term	²	df	p
F3	0.4446	1	0.5049
F7	0.2642	1	0.6072
GLOBAL	0.6022	2	0.7400

Global PH $p = 0.7400$ — no evidence of a PH violation.

D5 — YFB DeSurv + cohort:

Term	²	df	p
F7	2.1376	1	0.1437
F8	1.4036	1	0.2361
GLOBAL	4.9960	2	0.0822

Global PH $p = 0.0822$ — no evidence of a PH violation.

D2 — YFB orig (M5):

Term	²	df	p
F1	0.1265	1	0.7221
F2	0.0178	1	0.8940
GLOBAL	0.1588	2	0.9237

Global PH $p = 0.9237$ — no evidence of a PH violation.